

|                                                 |
|-------------------------------------------------|
| libero_90/10                                    |
| preposition +40 pp · p<0.0001                   |
| 100% place the black bowl on top of the cabinet |
| 60% place the black bowl up onto the cabinet    |

|                                     |
|-------------------------------------|
| libero_90/30                        |
| destination noun +56 pp · p<0.0001  |
| 76% put the black bowl on the plate |
| 20% put the black bowl on the dish  |

|                                              |
|----------------------------------------------|
| libero_90/31                                 |
| source color +20 pp · p=0.0226               |
| 36% put the bowl on top of the cabinet       |
| 16% put the black bowl on top of the cabinet |

|                                                          |
|----------------------------------------------------------|
| libero_90/38                                             |
| verb +20 pp · p=0.0440                                   |
| 54% pick up the right moka pot and place it on the stove |
| 34% lift the right moka pot and place it on the stove    |

|                                      |
|--------------------------------------|
| libero_90/44                         |
| verb +94 pp · p<0.0001               |
| 100% fire up the stove               |
| 6% turn on the stove                 |
| particle placement +14 pp · p=0.0374 |
| 20% turn the stove on                |
| 6% turn on the stove                 |

|                                                |
|------------------------------------------------|
| libero_goal/1                                  |
| destination noun +92 pp · p<0.0001             |
| 92% place the grey bowl on the electric burner |
| 18% place the grey bowl on the hotplate        |
| 0% place the grey bowl on the hot plate        |

|                                                     |
|-----------------------------------------------------|
| libero_goal/5                                       |
| source noun +60 pp · p<0.0001                       |
| 62% slide that plate over to the front of the stove |
| 2% slide that dish over to the front of the stove   |

|                               |
|-------------------------------|
| libero_goal/7                 |
| source noun +98 pp · p<0.0001 |
| 100% switch on the stove      |
| 86% switch on the range       |
| 78% switch on the hob         |

|                                   |
|-----------------------------------|
| 74% switch on the cooktop         |
| 66% switch on the heating element |
| 62% switch on the griddle         |
| 30% switch on the hotplate        |
| 2% switch on the hot plate        |
| verb +52 pp · p<0.0001            |
| 100% turn on the burners          |
| 48% start the burners             |

|                                    |
|------------------------------------|
| libero_goal/8                      |
| destination noun +52 pp · p<0.0001 |
| 100% put the bowl on the plate     |
| 48% put the bowl on the dish       |

|                                             |
|---------------------------------------------|
| libero_goal/9                               |
| destination noun +44 pp · p<0.0001          |
| 94% put the wine bottle on the rack         |
| 76% put the wine bottle on the stand        |
| 50% put the wine bottle on the wooden stand |

|                                                       |
|-------------------------------------------------------|
| libero_object/4                                       |
| modifier added +38 pp · p<0.0001                      |
| 96% pick up the bottle and place it in the basket     |
| 58% pick up the red bottle and place it in the basket |

|                                                   |
|---------------------------------------------------|
| libero_object/9                                   |
| source noun +84 pp · p<0.0001                     |
| 96% pick up the juice and place it in the basket  |
| 20% pick up the drink and place it in the basket  |
| 12% pick up the carton and place it in the basket |

|                                                                           |
|---------------------------------------------------------------------------|
| libero_spatial/1                                                          |
| locator noun +38 pp · p<0.0001                                            |
| 100% pick up the black bowl next to the ramekin and place it on the plate |
| 62% pick up the black bowl next to the dish and place it on the plate     |

|                                                                       |
|-----------------------------------------------------------------------|
| libero_spatial/7                                                      |
| locator noun +78 pp · p<0.0001                                        |
| 100% pick up the black bowl on the stove and place it on the plate    |
| 22% pick up the black bowl on the hot plate and place it on the plate |

Task scenes



libero\_90/10



libero\_90/30



libero\_90/31



libero\_90/38



libero\_90/44



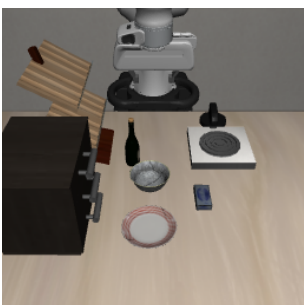
libero\_goal/1



libero\_goal/5



libero\_goal/7



libero\_goal/8



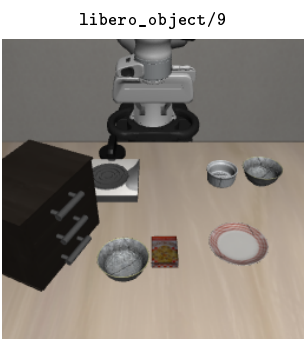
libero\_goal/8



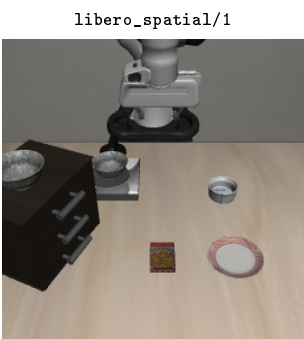
libero\_goal/9



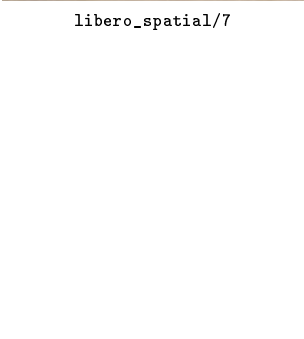
libero\_object/4



libero\_object/9



libero\_spatial/1



libero\_spatial/7